

Objective Apply Random Forest classification on the Heart Failure Clinical Records dataset to predict patient mortality (DEATH_EVENT) and evaluate model performance using a Confusion Matrix.	Algorithm Overview Random Forest builds multiple decision trees (ntree=100) and merges their majority-vote predictions. It reduces overfitting, handles imbalanced data well, and exposes feature importance via importance=TRUE.
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R Code Implementation

1. Load & Prepare

```
data <- read.csv(file.choose())
library("randomForest")
data$DEATH_EVENT <-
  as.factor(data$DEATH_EVENT)
```

2. Train / Test Split (80 / 20)

```
set.seed(123)
train_index <-
  sample(1:nrow(data),
        0.8 * nrow(data))
train_data <- data[train_index, ]
test_data <- data[-train_index,
]
```

3. Train Random Forest Model

```
model <- randomForest(
  DEATH_EVENT ~ .,
  data = train_data,
  ntree = 100,
  importance = TRUE)
print(model)
```

4. Predict & Accuracy

```
predictions <- predict(model,
  test_data)
accuracy <- sum(predictions ==
  test_data$DEATH_EVENT) /
  nrow(test_data)
print(accuracy)
```

5. Plot Confusion Matrix

```
library(ggplot2)
cm <- table(Predicted =
  predictions,
            Actual =
  test_data$DEATH_EVENT)
cm_df <- as.data.frame(cm)
ggplot(data = cm_df,
  aes(x=Actual, y=Predicted,
    fill=Freq)) +
  geom_tile(color = "black") +
  geom_text(aes(label=Freq),
    size=6) +
  scale_fill_gradient(
    low="lightblue", high="blue")
+
  labs(title="Confusion Matrix -
    Random Forest") +
  theme_minimal()
```

Confusion Matrix Output

Confusion Matrix

Random Forest Model Performance

Predicted \ Actual	0	1
1	6	12
0	36	6

Derived Metrics (from matrix)

Metric	Formula	Value
Accuracy	(TN+TP)/Total	80.0 %
Precision	TP/(TP+FP)	66.7 %
Recall	TP/(TP+FN)	66.7 %
F1-Score	2×P×R/(P+R)	66.7 %

Matrix Breakdown

Cell	Count / Meaning
TN (Pred=0, Act=0)	36 — Correctly predicted survived
TP (Pred=1, Act=1)	12 — Correctly predicted death
FP (Pred=1, Act=0)	6 — Predicted death, survived
FN (Pred=0, Act=1)	6 — Predicted survived, died

Conclusion

The Random Forest model (ntree=100, 80/20 split) achieved 80% accuracy on the Heart Failure test set. The confusion matrix shows 36 TN and 12 TP (correct predictions) against 6 FP and 6 FN (misclassifications). Balanced Precision and Recall of 66.7% suggest the model handles both classes reasonably, with scope for improvement via hyperparameter tuning or SMOTE oversampling for the minority death class.